

Rahul Saravanakumar

832-202-8910 | rahulsaravanakumar@tamu.edu | linkedin.com/in/rahulsaravanakumar | rhl.sh

EDUCATION

Texas A&M University

Aug. 2023 – May 2027

Bachelor's Degree in Computer Science, Minor in Business and Statistics

College Station, TX

- 3.81 GPA — Dean's List (4 Semesters)
- TA for Computer Science Program Design and Concepts
- Member of Aggie Coding Club and Aggie Competitive Coding Club

EXPERIENCE

Full Stack Developer

Dec. 2025 – Present

Otovo USA

Remote

- Designed and implemented an automated email and SMS delivery system.
- Built backend logic for user segmentation and segment-entry triggers to drive event-based email workflows.
- Adapted open source CMS enabling live content updates without requiring backend redeploys with agentic integrations
- Created cutting edge AI enabled integrations (MCP, bots, OpenClaw pods) to increase efficiency and ease of use.

Undergraduate Research Assistant - AI Image-Based Weed Detection

Jun. 2025 – Aug. 2025

Texas A&M University

College Station, TX

- Researched use of image segmentation models with YOLO for agricultural object detection, improving weed classification accuracy by 28%.
- Trained and evaluated deep learning models, achieving 82% mAP and 0.67 IoU on test datasets for multi-class weed detection.
- Collected and annotated over 2,000 field images to create a high-quality training dataset for weed identification.

PROJECTS

HEBpath | *Rust, WebAssembly, JavaScript*

Aug. 2025 – Present

- Built a fully client-side tool that computes the shortest in-store path for grocery cart item collection.
- Developed a Rust → WASM engine to map items onto a store SVG and run A* pathfinding on a rasterized grid.
- Created an iOS Shortcut pipeline to extract cart location data, compress it, and pass it via URL fragments.
- Rendered and cached optimized routes directly in-browser with no backend dependencies.

BeliNearMe | *Python, JavaScript, REST APIs, Geolocation*

Aug. 2025 – Present

- Reverse engineered Beli's mobile API requests to access hidden restaurant search query functionality.
- Extracted and repurposed endpoint parameters to find top-rated restaurants near a given location.
- Built a custom query layer to rank nearby restaurants by distance, popularity, cuisine, and review signals.
- Created iOS hook to automate the restaurant discovery flow to bypass manual in-app searching and surface better food options faster.

Music Retrieval AI Assistant | *Python, PyTorch, Pinecone, Spotify Web API*

Aug. 2025 – Present

- Built a Spotify-integrated assistant for semantic and acoustic song search from natural language queries.
- Finetuned a domain-specific LLM to extract structured search parameters from unstructured input with 93% production accuracy.
- Implemented a RAG-based dual-vector retrieval system using metadata and CLAP embeddings, reducing mismatches by 38%.
- Indexed 10K+ tracks in Pinecone with audio feature filters, improving mood-based search latency by 45%.

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++/C#, Java, Rust, TypeScript, SQL (PostgreSQL, SQLite3), R, Bash

Frameworks: React, NumPy, Pandas, Node.js, PyTorch, TensorFlow, Flask, FastAPI, Django, jQuery

Developer Tools: Git, AWS, Azure, Word, Excel, Tableau, Firebase, Docker, Google Cloud Platform, VS Code

Relevant Coursework: Data Structures & Algorithms, Processor Design, Operating Systems, Programming Languages, Machine Learning